Lecture – 1

❖ **Computer hardware and software**

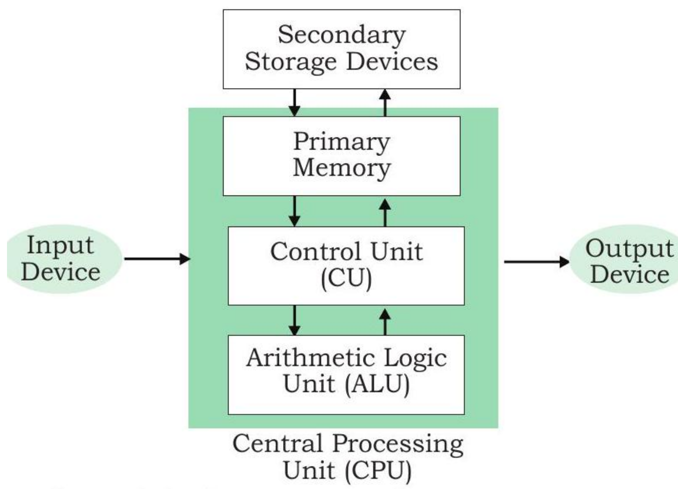
❒ **What is Computer?**

- A computer is an electronic device, operating under the control of instructions stored in its own memory that can accept data (input), process the data according to specified rules, produce information (output), and store the information for future use.

❒ **Functionalities of a Computer**

Any digital computer carries out five functions in gross terms.

✓ Takes Data as Input
✓ Stores the data/Instructions in its memory and use them when required.
✓ Processes the data and coverts it into useful Information.
✓ Generates the Output.
✓ Control all the above four steps.



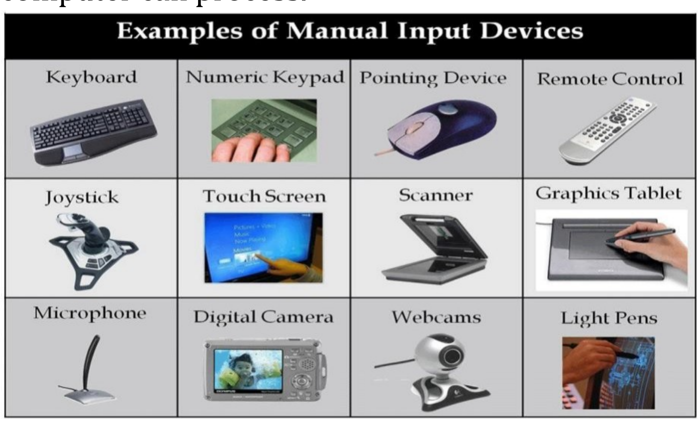
❒ **Computer Hardware**

- Computer hardware means the physical parts of a computer.
- Physical parts are things that we can see and touch.
- Some common hardware components are:
  - ***Monitor*** *– Displays information on the screen.*
  - ***Keyboard*** *– Used to type letters and numbers.*
  - ***Mouse*** *– Used to point, click, and select items.*
  - ***Hard Disk (HDD)*** *– Stores data and files.*
  - ***Memory (RAM)*** *– Temporarily stores data while programs are running.*
  - ***Motherboard*** *– Connects and allows different components to communicate.*
  - ***Graphics Card*** *– Helps display images and videos.*
  - ***Sound Card*** *– Helps produce and process sound.*
  - ***Processor (CPU)*** *– Performs calculations and controls computer operations.*

- All these physical components together make up the computer hardware.
- In simple words:
- ***Computer Hardware = All the physical parts of a computer that we can see and touch***.

➢ **Input Devices:**

- An input device helps us enter data and instructions into a computer.
- It converts information from a human-understandable form into a form the computer can process.



➢ **Central Processing Unit (CPU):**

- The **CPU is the brain of the computer**.
- It **processes data and executes instructions**.
- It controls and performs **most of the functions and operations** of the computer.
- The CPU is a **key component that determines the computing power and performance** of a computer.

The CPU is comprised of three main parts:

1. ***Arithmetic Logic Unit (ALU)***: Executes all arithmetic and logical operations. Arithmetic calculations like as addition, subtraction, multiplication and division. Logical operation like compare numbers, letters, or special characters

2. ***Control Unit (CU):*** Every Operation such as storing, computing and retrieving the data should be governed by the control unit.
3. ***Memory Unit (MU):*** The Memory unit is used for storing the data. The Memory unit is classified into two types.
   i. ***Primary Memory***

      RAM (Random Access Memory)
      - ✓ RAM is temporary memory used by the computer.
      - ✓ It stores data and programs that the CPU is currently using.
      - ✓ It is volatile, so data is lost when the power is turned off.
      - ✓ The CPU can quickly access any location in RAM.

      ROM (Read Only Memory)
      - ✓ ROM is permanent memory used by the computer.
      - ✓ It stores important instructions needed to start and operate the computer.
      - ✓ It is non-volatile, so data remains even when power is turned off.
      - ✓ Normally, the data stored in ROM cannot be easily modified.

   ii. ***Secondary Memory***

      Magnetic Storage:

      The Magnetic Storage devices store information that can be read, erased and rewritten a number of times.

      Example: Floppy Disks, Hard Disks, Magnetic Tapes

      Optical Storage:

      The optical storage devices that use laser beams to read and write stored data.

      Example: CD (Compact Disk),DVD(Digital Versatile Disk)

❒ **Computer Software**

- Software of a computer system can be referred as anything which we can feel and see.
  Example: Windows, icons
- Computer software is divided in to two broad categories: **system software** and **application software**.

- **System Software**
  - ✓ System software is software that controls and manages computer hardware and provides a platform for application software to run.
  - ✓ It acts as a bridge between hardware and the user/application software.
    Examples: Windows, Linux, macOS, Android etc.
- **Application software**
  - ✓ Application software is designed to help users perform specific tasks or activities.

- ✓ Unlike system software, application software is primarily designed for end users.
  Examples: MS Word, MS Excel, PowerPoint etc.
- ***Hardware and Software Relationship***
  - ✓ Hardware and software depend on each other.

USER

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APPLICATION SOFTWARE

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SYSTEM SOFTWARE

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HARDWARE

- ✓ For example, when playing a video:

  VLC Media Player → Application Software
  Windows/Linux → System Software
  CPU, RAM, GPU, Speakers → Hardware